MILITARY SPECIFICATION

LOADING PLATES, STEEL, AND ALUMINUM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three types of loading plates, three styles of aluminum plates and two styles of steel plates rated at a 15,000 pound load capacity. These loading plates will be used for bridging gaps and compensating for height differentials between docks and trucks, docks and railcars, and railcars and railcars.

1.2 Classification. Loading plates shall be of the following types and styles as specified (see 6.2):

Type I - Rectangular
Style A Aluminum
Style B Steel

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research Development and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC/N/A FSC 3990

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Type II - Flared Sides
Style A Aluminum
Style B Steel

Type III - Combination Loading Plate and Ramp
Style A Aluminum

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-A-200/8	- Aluminum Alloy Bar, Rod, Shapes, Tube and Wire, Extruded, 6061.
QQ-A-225/8	- Aluminum Alloy Bar, Rod Wire and Special Shapes; Rolled, Drawn, or Cold Finished 6061.
QQ-A-250/11	- Aluminum Alloy 6061, Plate and Sheet.

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MIL-T-704	- Treatment and Painting of Materiel.
MIL-P-15024	- Plates, Tags and Bands For Identification of Equipment.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-W-22248	- Weldments, Aluminum and Aluminum Alloy.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

FEDERAL

FED-STD-595	- Colors.

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1303	- Painting of Naval Ordnance Equipment.
MIL-STD-2175	- Castings, Classification and Inspection of.

(Copies of specifications and standards required by contractors in connectition with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government publications. The following other Government publications form a part of this specification to the extent specified herein.

DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR, Chapter XVII, Part 1910, Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 632 - Aluminum-Alloy Rolled Tread Plate, Standard Specification for.
D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the Amrtivsn Soceiety for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

AWS B2.1 - Standard for Welding Procedure and Performance Qualification.

(Application for copies should be addressed to the American Welding Society, 2501 N.W. 7th Street, Miami, FL 33125.)

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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The loading plates shall be rectangular, flared side, or combination loading plate and ramp type and of aluminum or steel as specified (see 1.2). The plates shall be of welded construction. Aluminum loading plates shall be equipped with hand grips (see 3.4.3). Steel loading plates and when specified (see 3.4.2), aluminum loading plates shall be equipped with lifting devices for handling by forklift truck to position the loading plates. Loading plates shall permit safe and efficient use of forklift trucks for loading and unloading operations. Load plate shall include commercial item designated as dock board.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Material for aluminum safety curbing, shall be in accordance with QQ-A-200/8 or QQ-A-250/11. Material for the underside structural members, shall be in accordance with QQ-A225/8 or QQ-A-200/8.

3.3.1 Material deterioration and control. The loading plates shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

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3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the loading plates may be newly fabricated from recovered materials to the maximum extent practicable, provided the loading plates produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the loading plates.

3.3.3 Castings. Castings provided on all style A loading plates shall be at least of the quality normally used in commercial practice, free from injurious blowholes, porosity, shrinkage, cracks, or other defects. The use of repaired castings in which the repairs could adversely affect the strength of the castings shall not be permitted. Castings shall be in accordance with MIL-STD-2175, class 3.

3.4 Design and construction. Loading plates shall be of welded construction, except for the retaining and lifting devices (chains with recessed hooks, and lock-stops), which may be secured by other means.

3.4.1 Structural design. Loading plates shall be designed with a safety factor of 200 percent of rated load capacity based on yield strength of the material. The underside structural members shall extend to at least 7.50 inches below the floor plate to insure positive blocking between door of truck or railcar and dock edge. The combination loading plate and ramp shall be designed such that the two units shall lock together with locking pins, to form one unit. The locking pin position shall be protected by safety curbing. Ramplocks shall be provided to stabilize the loading plate. Ramplocks shall be of adjustable type and shall be attached to the ramp section of type 111 loading plates.

3.4.2 Lifting devices. Style B loading plates shall have lifting devices so that the loading plates may be transported and positioned by a forklift truck having forks 36 inches long, 6 inches wide and located on 24 inch centers. The style A loading plates shall, when specified (see 6.2), have lifting loops with dimensions not less than 7 inches wide and 3 inches high on 26 inch centers. The loops shall drop by gravity flush with the floor plate when not in use. Lifting loops are to be removable for maintenance. Removal of loops shall be accomplished with simple hand tools.

3.4.3 Hand grip. Style A loading plates shall be equipped with continuous hand grips located along each side or with four hand grips, two located within each of the side curbs on the outside, near the ends. Hand grip holes are to be a minimum of 5 inches long X 2.50 inches wide and 2 inches deep. Hand grips shall be free of sharp edges and sealed to prevent accumulation of moisture within the side curbs.

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3.4.4 Safety curbing. Loading plates shall have safety curbing along the full length of each side to prevent forklift trucks from going over the sides. The curbing shall have all edges and corners rounded. Side curbs shall have flared ends permitting sharp angle entry and exit. The curb ends shall be curved outward and downward. Side curb height shall be a minimum of 2 inches above the loading plate top floor surface, and shall be designed to minimize wheel rim/curb contacts. The width of the side curbs at the base shall not reduce the usable width of the loading plate by more than 6 inches. The side curbs shall be painted a. yellow in accordance with system 76 of MIL-STD-1303, color yellow gloss No. 13538 of FED-STD-595.

3.4.5 Floor surface. The floor surface shall have a diamond-pattern, skid-resistant tread pattern. Both ends of the loading plate floor surface shall be beveled along the entire width to provide smooth entry of the forklift truck. Edge thickness shall be 0.25 inch maximum and 0.18 inch minimum. Style A loading plates shall have floors in accordance with ASTM B632, alloy 6061-T6, except that the thickness shall be 0.625 inch minimum, not including the thickness of raised pattern.

3.4.6 Position lockstop. Types I and III loading plates shall be provided with a fixed position lockstop located on the underside of the floor surface. The position lockstop shall act as a positive stop in both directions under conditions of 0 to plus 6 inches height differential between the loading platform end and the truck end. The lockstop shall allow truck park-out within 2 inches.

3.4.7 Retaining devices, type II. Type 11 loading plates shall be provided with a fixed position lockstop and an adjustable retaining device. The retaining device shall be adjustable in 2 inch increments in a range between 10 inches and 42 inches. The lockstop and retaining device shall act as a positive stop in both directions under conditions of 0 to plus 6 inches height differential between the loading platform end and the railcar end.

3.5 Performance.

3.5.1 Load capacity. The loading plates shall have stable operation at a rated load of 15,000 pounds. The loading plates shall show no permanent deformation or failure when subjected to a test load of 200 percent of axle load capacity when tested as specified in 4.5.2.2.

3.5.2 Safe operation. The loading plates shall meet or exceed current OSHA requirements.

3.6 Dimensions. Unless otherwise specified (see 6.2), loading plates shall conform to the overall dimensions specified in table I.

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TABLE I. Dimensions.

	Type I (inches)	Type II (inches)	Type III (inches)
Width at platform end (overall)	64	96	64
Width at railcar or truck end (overall)	64	70	64
Length (overall)	44	48	44

Note: A tolerance of plus or minus 1 inch shall be allowed on the above dimensions.

3.7 Identification marking. Each loading plate shall be marked in accordance with MIL-STD-130. Each loading plate shall have an identification plate conforming to MIL-P-15024, type A. The plate will be securely attached in an protected location. The following information shall be permanently marked, as applicable, on the name plate in characters not less than 0.375 inches:

Loading Plate
MIL-L-40064, Type____ Style _____
Company Name
Capacity: _____ pounds

3.8 Treatment and painting. The style B loading plates shall be cleaned, treated, and painted in accordance with MIL-T-704, type A, color No. 13538 of FED-STD-595. Style A loading plates shall be cleaned only.

3.9 Workmanship. All loading plates shall be clean and properly assembled. Burrs shall be removed and sharp edges shall be rounded. Parts shall not be damaged or impaired in any way, and parts shall show no evidence of rust or corrosion.

3.9.1 Welding. All surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. parent materials. weld filler metals, and fabrication techniques shall be as required to enable the loading plate to conform to the examination and test requirements specified in section 4. Parts to be joined by welds shall be brought into as close contact as possible and in no event shall be separated by more than 0.18 inch unless appropriate bridging techniques are used. The welding process used in fabrication shall be at the option of the contractor. Aluminum alloy welding and inspection shall be in accordance with MIL-W-22248, class 3. Welding shall be inert gas shielding. Welds shall be inspected for good adhesion, no porosity and good penetration.

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3.9.2 Qualifications for welders and welding equipment operators. The contractor shall be responsible for qualifying all welders and welding equipment operators in accordance with AWS 32.1 using the contractor's or manufacturer's recorded welding procedure. The Government may require the recertification of any welder or welding equipment operator.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, inspected, and tested in accordance with applicable specifications and standards in this specification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article loading plate shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article loading plate shall be tested as specified in 4.5.2.1 and 4.5.2.2. Failure of any test shall be cause for rejection.

4.4 Quality conformancxw inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105. A lot shall consist of all loading plates of one type and style offered for delivery at one time.

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4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.1, 4.5.2.2 and 4.5.2.3. Inspection level shall be S-3 with AQL of 1.0 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The sample loading plate shall be examined as specified herein for the following defects:

Major

Defect No.	Defect	Requirement Paragraph
101.	Loading plates not as specified.	3.1
102.	Materials not as specified.	3.3
103.	Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.	3.3.1
104.	Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.	3.3.1.1
105.	Contractor does not have documentation available for identification of material, material finishes, or treatment.	3.3.1.2
106.	Used, rebuilt or remanufactured components, pieces, or parts incorporated in the loading plate.	3.3.2
107.	Castings not as specified.	3.3.3
108.	Construction not as specified.	3.4
109.	Structural design not as specified.	3.4.1
110.	Lifting devices not as specified.	3.4.2
111.	Handgrips not as specified.	3.4*3
112.	Safety curbing not as specified.	3.4.4
113.	Floor surface not as specified.	3.4.5
114.	Position lockstop not as specified.	3.4.6
115.	Retaining devices, type IX not as specified.	3.4.7
116.	Dimensions not as specified.	3.6
117.	Workmanship not as specified.	3.9
118.	Welding not as specified.	3.9.1

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Minor

201.	Identification marking not in accordance with MIL-STD-130.	3.7
202.	Treatment and painting not in accordance with MIL-T-704, type A, color No. 13538.	3.8

4.5.2 Tests.

4.5.2.1 Operational. The loading plates shall be positioned with a height differential of zero inch and 6 inches with the railcar or truck end at the higher level. The span opening shall be 2 inches or less for types I and III loading plates and 10 inches, 26 inches, and 42 inches for type II loading plates, when lifting loops are specified, and style B loading plates shall be lifted, transported and positioned, utilizing the lifting loops, by a forklift truck having forks 36-inches long, 6-inches wide and located on 24-inch centers. At each height position with each span opening, a forklift truck having a gross vehicle weight (GVW) of 15,000 pounds shall be driven forward across the plate from the dock to the railcar or truck and then backward from railcar or truck to the dock. Nonconformance to 3.4.2, 3.4.3, 3.4.6, and 3.4.7 as applicable, or evidence of instability shall constitute failure of the plate.

4.5.2.2 Static load. The loading plate shall be placed over a span of not less than 20 inches for type I and not less than 42 inches for type II loading plate. Type III loading plate shall be placed over a span of not less than 20 inches across the dockboard section. The platform end and the truck or railcar end shall be supported at the same height. The loading plate shall be subjected to a static load of 30,000 pounds applied over an area extending not more than 2 inches on each side of the transverse center-line of the loading plate for the full width of the loading plate. The load shall remain on the loading plate for a period of 30 minutes, after which the load shall be removed and the loading plate shall be inspected. Evidence of permanent deformation, weld failure or damage shall constitute failure of this test. During this overload test the loading plate shall be supported as during normal use.

4.5.2.3 Lifting loops. Lifting loops, when provided, shall be tested as follows. The loading plate shall be supported by a 4 inch wide forklift tine or equivalent support member through each lifting loop. The support members shall be located centrally within the lifting loops. The dock board shall be loaded with twice the weight of the loading plate while being supported as described above. This overload shall remain for 5 minutes. Permanent deformation, weld failure, or damage of any kind shall constitute failure of this test.

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4.6.1 First article pack inspection. The first article pack shall be examined for. the defects specified in 4.6.2.3. Presence of one or more defects shall be cause for rejection.

4.6.2 Quality conformance inspection of pack.

4.6.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following defects. The AQL shall be 2.5 percent defective.

No.	Defect	Paragraph
119.	Loading plates not preserved and packed in a manner to assure carrier acceptance.	5.1
120.	Loading plates not preserved and packed in in a manner to assure safe delivery to destination at lowest traffic rates.	5.1
121.	Loading plates not marked as specified.	5.2

5. PACKAGING

5.1 Preservation and packing. The loading plates shall be preserved and packed in accordance with the requirements of ASTM D 3951 in a manner to assure carrier acceptance and safe delivery to destination at lowest traffic rates.

5.2 Marking. In addition to any special or other identification marking required by the contract or purchase order, and unless otherwise specified (see 6.2), the loading plates shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Type I loading plate is intended primarily for truck operations, type II loading plate is intended primarily for railcar operations, and type III is intended primarily for use when height differences between dock and truck or dock and railcar are extreme. The style A loading plates covered by this specification are intended for use in dock-to-railcar, or railcar-to-railcar, or dock-to-truck operations. The style B loading plates covered by this specification are intended to be used to span openings between loading platform and entrances to railcar or trucks. The loading plates can be safely used to support forklift trucks carrying full capacity loads with maximum gross vehicle weight GVW of 15,000 pounds.

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type loading plate required (see 1.2).

c. When a first article is required for inspection and approval and the number of units required (see 3.2 and 6.3).

d. Quantity of each type required..

e. men lifting loops are required (see 3.4.2).

f. When dimension is other than specified (see 3.6).

g. When marking is other than specified (see 5.2).

h. Any special marking (see 5.2).

6.3 First article. When a first article inspection is required, the items should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval or the first article test results and disposition of the documents first articles.

6.4 Levels of preservation and packing have not been included as the nature of the item is such that only the single method specified is appropriate. Preservation and packing level markings shall be appropriate to the type of shipment and storage involved.

Custodians:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-40064E	Loading Plates, Steel, and Aluminum

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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